

Tingjian (Terence) Yu

Waterloo, ON | 548-398-2250 | terenceyu@gmail.com | [LinkedIn](#) | [Website](#)

Profile

Senior software engineer with over 13 years of experience in designing and delivering scalable, cloud-native applications built on event-driven microservices serving millions of clients. Adept at optimizing system performance and reliability through robust design and maintaining clean, testable code.

Work Experience

Software Engineer

Feb 2025 – Jun 2025

Outlier, Canada

- Reviewed AI-generated solutions in JavaScript, Go, and Python, validating unit tests and edge cases to ensure production-grade quality, scalability, and adherence to best practices, collaborating with AI-assisted development workflows.

Senior Backend Engineer

Dec 2017 – Mar 2023

Ecovacs Robotics, China

- Led a group of 8 to build over a dozen backend applications supporting over 10 million users and devices, handling 30,000 peak RPS, including NLP systems, IoT platforms, image processing, and smart home automation using Go, Node.js, and Python, combining Express, Gin, FastAPI, Protobuf, REST, PostgreSQL, Kafka, and Redis.
- Designed architectures for scalability, security, and 99.99% high availability based on event-driven microservices, enhanced API design with Swagger, and relational modeling, incorporating asynchronization, caching, connection pooling, query optimization, and tools like Prometheus, pprof, and Elastic APM to analyze and improve systems.
- Optimized performance through refactoring from monolith to microservices, reducing interface latency by 90% and achieving P95 under 300ms by rewriting Node.js with Go, Goroutine, CGo, and OpenCV, and offloading CPU-intensive computation from server to edge devices via a cross-compiled dynamic library.
- Created a smart home automation platform handling millions of daily API calls, integrated with a secure website enabling third-party developers to register applications, configure permissions, and execute operations via JWT and REST APIs, using Python, Pytest, React, Jest, Playwright, AWS Lambda, MongoDB, and OAuth 2.0.
- Built a voice assistant by modularly integrating third-party AI interfaces and architected an annotation system to increase the speech recognition rate by 7% through training customized speech models, using streaming gRPC, S3, Azure, and GCP.
- Streamlined software release processes by utilizing CI/CD tools like Git, Jenkins/GitHub Actions, SonarQube, and automated testing, shortening deployment time by 2/3, and reducing errors by 70%.
- Partnered with PM/UX/Firmware teams in 2-week sprints to prioritize features, enabling on-time delivery per sprint.
- Mentored 3 junior engineers through code reviews, reducing onboarding time from 3 months to 6 weeks.

iOS Developer

May 2012 – Dec 2017

Ecovacs Robotics, China

- Built the company's first robot-controlling mobile application with OC, Swift, and Cordova from scratch, used by over 100K users, encapsulating IoT communicating SDK and participating in the definition of prototypes and UX/UI.

Embedded Software Developer

Mar 2010 – May 2012

Qisda Corporation, China

- Built firmware and driver programs of optical sensor and motor with C and Assembly according to the chip's spec.

Skills

Languages: Go, JavaScript / TypeScript, Python, Java, C, HTML.

Backend Development: Network Programming, Distributed Systems, Microservices, Design Patterns, REST, gRPC, GraphQL, OAuth, PostgreSQL, MySQL, MongoDB, Cassandra, ClickHouse, Redis, Kafka, RabbitMQ, S3.

Cloud and Tools: OpenAPI, Git, Agile, AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Shell Scripting, Linux, Elastic Stack, Prometheus, Grafana, Terraform.

Education

Postgraduate Certificate of Web Development (Distinction) | *Conestoga College*

2023 – 2024

Bachelor of Electronic Information Engineering (Academic Scholarship) | *Chongqing University*

2005 – 2009